Statistical Analysis for Grass Data

The experiment for this data set is described in the SAS file for Grass growth experiment data, that is posted on Husky CT. The statistical model for this experiment is:

$$Y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + \epsilon_{k(ij)},$$

where $i = 1, 2, 3,\ j = 1, 2, 3, 4, 5$ and $k = 1, 2, 3, 4, 5, 6$. The factor $A =$ METHOD and $B =$ VARIETY. Both factors A and B are fixed factors. Based on the description of the experiment this is a completely randomized two factorial design, with 6 replicates. Note that the data is given in the following form: in each row, for each combination of the levels of factor A and B, the yield for the 6 replicates are listed. That is why the following INPUT statement had to be used:

```
INPUT METHOD $ VARIETY @;
      DO REPLICATE = 1 TO 6;
                INPUT YIELD @;
                      OUTPUT;
                             END;
```

The \$ sign after METHOD indicates that the levels of METHOD are not numeric. The INPUT statement lists the levels of METHOD and VARIETY and DO loop inputs the six values for the response variable YIELD, that is defined by the second INPUT statement. Note that each INPUT statement ends with @ symbol. Since there was only one DO loop statement, we have one OUTPUT and one END statement, both ending with a semicolon

In this experiment we are interested to test simultaneously the following three null hypotheses:

$$H_{01} : A_i = 0, 1 \leq i \leq 3, \tag{1}$$

$$H_{02} : B_j = 0, 1 \leq j \leq 5, \tag{2}$$

and

$$H_{03} : (AB)_{ij} = 0, 1 \leq i \leq 3, 1 \leq j \leq 5. \tag{3}$$

The test statistic for testing the null hypothesis $H_{01} : A_i = 0, 1 \leq i \leq 3$ is:

$$F_1 = \frac{MS_A}{MS_E},$$

the test statistic for testing $H_{02} : B_j = 0, 1 \leq j \leq 5$ is:

$$F_2 = \frac{MS_B}{MS_E},$$

and the test statistic for testing $H_{03} : (AB)_{ij} = 0, 1 \leq i \leq 3, 1 \leq j \leq 5$ is:

$$F_3 = \frac{MS_{AB}}{MS_E}.$$

Suppose we decided to use an overall significance level of $\alpha = .10$, which means that the nominal significance level for each test is .033. If the underlying assumptions for our model are valid, then it follows from page 2, Table for Type III SS in the SAS output for grass growth experiment data file that:

$$F_1 = 24.25,\ p - value < .0001,$$

$$F_2 = 0.14,\ p - value = 0.9648,$$

and

$$F_3 = 2.38,\ p - value = 0.0241.$$

Since the p-values associated with the test statistics for testing H_{01} and H_{03} are less than .033, we reject these hypotheses. The p-value associated with the test statistic for testing H_{02} is greater than .033, we cannot reject this hypothesis. Hence, the conclusion is that the METHOD has a significant effect of the YIELD of grass, the VARIETY factor by itself has no effect of the YIELD of grass, but the interaction between METHOD and VARIETY has a significant effect on grass.

Now before continuing with appropriate multiple comparisons of means of YIELD, let us examine the underlying assumptions for this model. From the Fit Diagnostics for Yield panel of plots, on page 3 in the SAS output for grass growth experiment data file we see that the plot of the Residual vs. Predicted Value has no patterns, so it looks like the variances of the random errors are stable and the linear model is adequate. The plot of RStudent vs. Predicted Value has no observations larger than 4 or smaller than - 4, implies that there are no outliers in the data. The plot of Cook's D values are all less than 1, which suggests that there are no influential observations. The plot of Residual vs. Quantile.is pretty straight, and the p-value of the W statistic, on page 14 of the SAS output, is equal to $.4772 > .10$, this confirms that the normality of the assumption for the random errors is valid This confirms that all the underlying assumptions for the mode are valid.

Since we have rejected H_{01} and H_{03} and accepted H_{02}, we have to perform the following Tukey multiple comparisons for the means of YIELD: we have to compare **simultaneously** the means of YIELD for the three levels of METHOD for each of the five levels of VARIETY. This implies that in the Tukey yardstick formula we have to use $a = 15$. Unfortunately, SAS does not do it for us and we have to use a calculator to compute the Tukey yardstick! The details are presented in the file "Multiple comparisons for grass growth experiment data, posted on Husky CT cite, along with the conclusions for this experiment.

Here is the Summary of the conclusions for this experiment:

Based on the multiple comparisons marked on the interaction plot of Method vs. Variety, one can conclude:

1. For Varieties 1, 2 and 3 methods A, B, C have not performed significantly different from each other.

2. For Variety no 4, Method A performed significantly better than Methods B and C. Methods B and C have not performed significantly different from each other.

3. For Variety no 5, Method A performed significantly better than Method C. The other two pairwise comparisons have not shown significant differences.